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EX. NO.10

TROUBLESHOOTING FABRIC APPLICATIONS

AIM

To identify, analyze, and resolve common issues encountered in Hyperledger Fabric applications by troubleshooting network connectivity, chaincode deployment, channel configuration, certificate authentication, and Docker container errors.

SOFTWARE REQUIREMENTS

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

PROCEDURE

Step 1: Navigate to the Fabric Test Network

Navigate to the Hyperledger Fabric test network directory. cd ~/fabric-dev/fabric-samples/testnetwork

Step 2: Start the Hyperledger Fabric Network

Stop any existing network and start a fresh Fabric network with the application channel and Certificate Authorities.

./network.sh down

./network.sh up createChannel -ca

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ cd ~/fabric-dev/fabric-samples/test-network
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -ca
Using docker and docker compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with crypto from 'Certificate Authorities'
Bringing up network
```

Step 3: Verify Docker Containers

Check whether all required Fabric containers are running.

docker ps

Verify the peer nodes, orderer, and Certificate Authority containers.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID   IMAGE                               COMMAND                  CREATED          STATUS          PORTS                                                                                                NAMES
2c5baef255f3   hyperledger/fabric-peer:latest      "peer node start"        31 seconds ago   Up 30 seconds   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp   peer0.org2.example.com
a44b18b6714e   hyperledger/fabric-orderer:latest   "orderer"                31 seconds ago   Up 30 seconds   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   orderer.example.com
ab1b97191d8a   hyperledger/fabric-peer:latest      "peer node start"        31 seconds ago   Up 30 seconds   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp   peer0.org1.example.com
14b030542e7e   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"   45 seconds ago   Up 44 seconds   0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp   ca_org1
818d5fea0349   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"   45 seconds ago   Up 44 seconds   0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp, 7054/tcp, 0.0.0.0:19054->19054/tcp, [::]:19054->19054/tcp   ca_orderer
9d18e23bd759   hyperledger/fabric-ca:latest        "sh -c 'fabric-ca-se…"   45 seconds ago   Up 44 seconds   0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp, 7054/tcp, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp   ca_org2
```

Step 4: Check Peer Logs

Display recent logs from the Org1 peer.

docker logs --tail 50 peer0.org1.example.com

Use the logs to identify peer startup, communication, channel, and chaincode-related problems.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs --tail 50 peer0.org1.exa
mple.com
2026-08-31 16:48:12.384 UTC 001c INFO [discovery] NewService -> Created with config TLS: true,
 authCacheMaxSize: 1000, authCachePurgeRatio: 0.750000
2026-08-31 16:48:12.384 UTC 001d INFO [nodeCmd] serve -> Discovery service activated
2026-08-31 16:48:12.384 UTC 001e INFO [nodeCmd] serve -> Starting peer with Gateway enabled
2026-08-31 16:48:12.384 UTC 001f INFO [nodeCmd] serve -> Starting peer with ID=[peer0.org1.exa
mple.com], network ID=[dev], address=[peer0.org1.example.com:7051]
2026-08-31 16:48:12.386 UTC 0020 INFO [nodeCmd] serve -> Started peer with ID=[peer0.org1.exam
ple.com], network ID=[dev], address=[peer0.org1.example.com:7051]
2026-08-31 16:48:12.387 UTC 0021 INFO [kvledger] LoadPreResetHeight -> Loading prereset height
 from path [/var/hyperledger/production/ledgersData/chains]
2026-08-31 16:48:12.387 UTC 0022 INFO [blkstorage] preResetHtFiles -> No active channels passe
d
2026-08-31 16:48:18.742 UTC 0023 INFO [ledgermgmt] CreateLedger -> Creating ledger [mychannel]
 with genesis block
2026-08-31 16:48:18.783 UTC 0024 INFO [blkstorage] newBlockfileMgr -> Getting block informatio
```

Step 5: Check Orderer Logs Display recent logs from the ordering service. docker logs --tail 50 orderer.example.com

Use the logs to identify ordering-service and block-processing issues.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs --tail 50 orderer.exampl
e.com
 Version: v2.5.16
 Commit SHA: f871cf9
 Go version: go1.26.4
 OS/Arch: linux/amd64
2026-08-31 16:48:11.755 UTC 000f INFO [orderer.common.server] Main -> Beginning to serve reque
sts
2026-08-31 16:48:15.437 UTC 0010 INFO [blkstorage] newBlockfileMgr -> Getting block informatio
n from block storage
2026-08-31 16:48:15.466 UTC 0011 INFO [orderer.consensus.etcdraft] HandleChain -> EvictionSusp
icion not set, defaulting to 10m0s
2026-08-31 16:48:15.466 UTC 0012 INFO [orderer.consensus.etcdraft] HandleChain -> Without syst
em channel: after eviction Registrar.SwitchToFollower will be called
2026-08-31 16:48:15.467 UTC 0013 INFO [orderer.consensus.etcdraft] createOrReadWAL -> No WAL d
ata found, creating new WAL at path '/var/hyperledger/production/orderer/etcdraft/wal/mychanne
l' channel=mychannel node=1
```

Step 6: Verify Channel Membership Load the Org1 environment if required. source setOrg1.sh

List the channels joined by the peer.

peer channel list

Verify that mychannel is available.

Step 7: Verify Installed Chaincode

Check the chaincode packages installed on

the peer. peer lifecycle chaincode

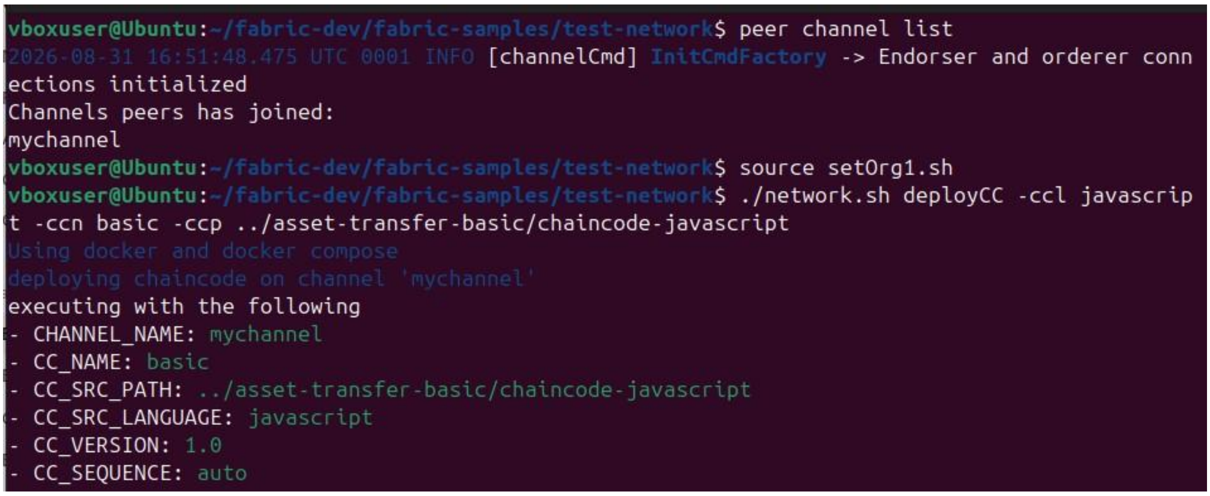
queryinstalled

This helps determine whether the required chaincode package has been installed correctly.

Step 8: Troubleshoot Chaincode Deployment

If the chaincode has not been installed or committed correctly, redeploy it.

./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transferbasic/chaincode-javascript



Step 9: Verify Chaincode Execution

Query the ledger to verify that the chaincode is functioning correctly. peer

chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7
050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -
n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:
9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"InitLedger","Args":[]}'
2026-08-31 16:53:30.660 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invok
e successful. result: status:200
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n
 basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset
"},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"
},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"a
sset"},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":
"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docT
ype":"asset"},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"
docType":"asset"}]
```

Step 10: Check Docker Resource Usage

Monitor CPU and memory usage of the Fabric

containers. docker stats

Allow the statistics to run for a few seconds and press Ctrl+C to

stop. For a single resource-usage snapshot:

docker stats --no-stream

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker stats --no-stream


CONTAINER ID   NAME                        CPU %     MEM USAGE / LIMIT     MEM %     NET I/O
      BLOCK I/O         PIDS
2c5baef255f3   peer0.org2.example.com    7.26%     26.55MiB / 5.625GiB   0.46%     225kB / 154k
B      45.1kB / 393kB   9
a44b18b6714e   orderer.example.com       0.39%     16.38MiB / 5.625GiB   0.28%     60.6kB / 207
kB     5.1MB / 193kB    8
ab1b97191d8a   peer0.org1.example.com    8.96%     30.62MiB / 5.625GiB   0.53%     225kB / 154k
B      0B / 393kB       8
14b030542e7e   ca_org1                   0.00%     9.527MiB / 5.625GiB   0.17%     40.1kB / 50k
B      0B / 737kB       7
818d5fea0349   ca_orderer                0.00%     11.05MiB / 5.625GiB   0.19%     60.5kB / 84.
6kB    999kB / 1.05MB   7
9d18e23bd759   ca_org2                   0.00%     9.633MiB / 5.625GiB   0.17%     37.9kB / 44k
B      1.16MB / 737kB   7
```

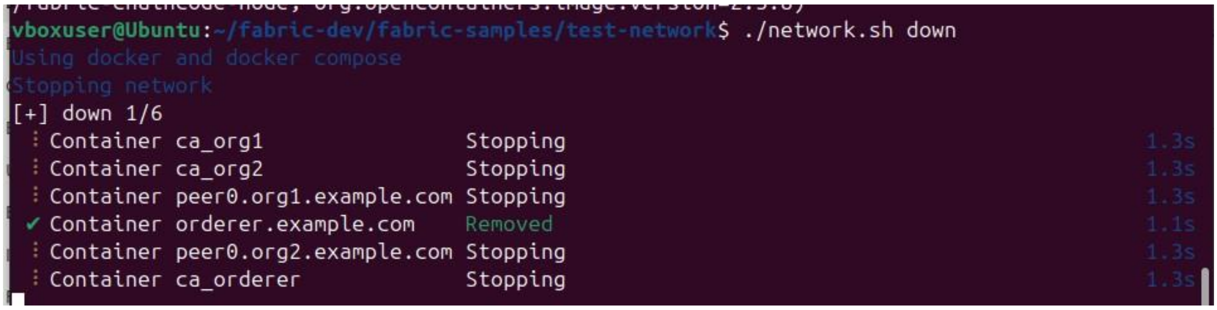
Step 11: Restart the Fabric Network

If network components are not responding correctly, restart the Fabric network.

./network.sh down

Start the network

again.



./network.sh up createChannel -ca

Step 12: Verify the Network After Restart

Check the running containers. docker

ps Verify channel membership. peer

channel list

Step 13: Clean Docker Resources

Optionally remove unused Docker

resources. docker system prune -f

This can be used to remove unused containers, networks, images, and other Docker resources.

COMMON ERRORS AND SOLUTIONS

Error	**Possible Cause**	**Solution**
Docker daemon not running	Docker service stopped	Start Docker using sudo systemctl start docker
Peer connection failed	Peer container stopped	Restart the Fabric network
Chaincode deployment failed	Incorrect chaincode path	Verify the -ccp directory
Channel not found	Channel was not created	Run ./network.sh createChannel
Access denied	Invalid MSP or certificates	Verify the MSP configuration and certificates

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Chaincode not installed Installation failed Redeploy the chaincode

TLS handshake failed Incorrect TLS certificate Verify the TLS CA certificate paths

Port already in use Another service is using the port

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Verify Channel

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TROUBLESHOOTING WORKFLOW

Start Fabric Network

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Verify Docker Containers

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Check Peer Logs

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Check Orderer Logs

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Verify Chaincode

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Test Application

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Identify Error

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Apply Solution

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Restart Network if Required

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Verify Successful Execution

RESULT

The Hyperledger Fabric application was successfully analyzed and common operational issues were identified and resolved using Docker commands, Fabric CLI tools, and log analysis. The experiment demonstrated effective troubleshooting techniques for network connectivity, chaincode deployment, channel configuration, certificate management, and resource monitoring.